

MAYANK SHRIVASTAVA

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Education

University of Illinois at Urbana-Champaign

Ph.D. in Computer Science

Specialization : Generative Models

Aug. 2024 – 2028 (Expected)

Illinois, USA

University of Illinois at Urbana-Champaign

Master of Science (Thesis-Track) in Computer Science

Specialization : Machine Learning, GPA: 4.0/4.0

Aug. 2022 – May 2024

Illinois, USA

Indian Institute of Technology Kanpur

Bachelor of Technology, Electrical Engineering with minor in Machine Learning

GPA: 9.8/10, Department Rank : 2/135

Aug. 2016 – Aug. 2020

Kanpur, India

Publications

Fisher Adjoint Matching: Natural Gradients for Stochastic Optimal Control

Mayank Shrivastava, Chandni Nagda, Rohan Deb, Arindam Banerjee

under review at UAI 2026



Flow Proposal Particle Filters for Data Assimilation

Chandni Nagda, Mayank Shrivastava*, Gudrun Thorkelsdottir, Arindam Banerjee*

under review at ICML 2026

Beyond Johnson–Lindenstrauss: Uniform Bounds for Sketched Bilinear Forms

Rohan Deb, Qiaobo Li, Mayank Shrivastava, Arindam Banerjee

AISTATS (Spotlight) 2025



Sketching for Distributed Deep Learning: A Sharper Analysis

Mayank Shrivastava, Berivan Isik, Qiaobo Li, Sanmi Koyejo, Arindam Banerjee

38th Annual Conference on Neural Information Processing Systems, 2024 (NeurIPS 2024)



Grand Theft Automation: Using VLMs & Algorithmic Prompt Optimization for Theft Detection

Mayank Shrivastava, John Mckay

GenAI Workshop on Advanced Techniques, Amazon Machine Learning Conference (AMLC 2025)

Max-Quantile Grouped Infinite-Arm Bandits

Ivan Lau, Yan Hao Ling, Mayank Shrivastava, Jonathan Scarlett

34th International Conference on Algorithmic Learning Theory, 2023 (ALT 2023)



Research Experience

*Equal contribution

Flow Proposal Particle Filters (FPPF)

Oct 2025 – Jan 2026

Research Assistant, advised by Prof. Arindam Banerjee

UIUC

- Proposed **FPPF**, a generative-model-based proposal for particle filtering in data assimilation.
- Designed SMC proposals that steer particles toward observation-consistent regions, reducing particle degeneracy.
- Implemented experiments on chaotic systems demonstrating improved filtering accuracy over autoregressive baselines.

Fisher Adjoint Matching

Oct 2025 – Feb 2026

Research Assistant with Dr. Arindam Banerjee

UIUC

- Proposed **Fisher Adjoint Matching (FAM)**, a natural-gradient method for fine-tuning flow-matching models.
- Identified instability in Adjoint Matching by showing it performs descent on a surrogate objective.
- Derived **KL-constrained updates** that guarantee monotonic improvement of the stochastic optimal control objective.
- Implemented experiments demonstrating improved stability and sample efficiency on control and sampling benchmarks.

Algorithmic Prompt Optimization for VLM-based Anomaly Detection

May 2025 – Aug 2025

Applied Science Summer Internship, Networks and Data Science Group

Amazon

- Developed a VLM-based change detection framework for automatic anomaly detection in warehouse photos.
- Built a two-stage pipeline using Claude VLM to describe visual changes and a text-based classifier to detect anomalies.

- Designed a prompt optimization (PO) framework where a white-box LLM generates candidate prompts.
- Applied surrogate optimization with a Neural UCB-based bandit method, improving AUC from 0.70 to 0.76.

Breaking the Dimension Dependence in Sketching for Distributed Learning July 2023 – March 2024

Research Assistant with Dr. Arindam Banerjee

UIUC

- Analyzed the convergence rates and communication efficiency of federated learning with linear sketching.
- Existing works inherit a dependence on ambient dimensionality - the size of the deep learning model.
- For overparametrized models, we derive dimension-free convergence rates for sketching-based federated learning.
- Extended our work to differentially private settings and heterogeneous clients, showing improvements in communication.

Max-Quantile Grouped Infinite-Arm Bandits

October 2020 – March 2022

Research Assistant with Dr. Jonathan Scarlett

NUS, Singapore

- Studied the problem of max-min grouped multi-arm bandits and extended it to the case of infinite arms.
- Developed a two-phase algorithm and derived instance-dependent sample complexity bounds.

Professional Experience

AI R&D Lab, Samsung Electronics

October 2020 – March 2022

Machine Learning Engineer, Automatic Speech Recognition team

Suwon, South Korea

- Worked on training and inference engine of Conformer Speech Model for Bixby's(virtual assistant) End-to-End ASR.
- Implemented neural model for External Language Model selection (shallow fusion) - improving inference accuracy by 3%
- Implemented RIR (Impulse Response) Augmentation for improving the accuracy on far-field ASR for IoT devices.

AI R&D Lab, Samsung Electronics, South Korea

May 2019 – July 2019

Summer Internship, Automatic Speech Recognition(ASR) Team

Suwon, South Korea

- Augmented female speech dataset by implementing Voice Conversion(VC) generative models on male speech dataset.
- Implemented generative models, including Conditional VAE, and Star-GAN VC in Pytorch- improving WER(Word Error Rate) of ASR engine by 2% for female speakers through the proposed augmentation pipeline.
- Received Pre-Placement Offer from Samsung as a result of exceptional performance and sincere effort.

Indian Institute of Science, Bangalore

May 2020 – Sep. 2020

Project Assistant with Dr. Himanshu Tyagi

Bangalore, India

- Developed an agent-based simulator for modeling COVID-19 spread in Karnataka, India.
- Modelled neighborhood movement as a Markov Chain using OpenCV to extract geographical location using Maps.
- Developed sampling-based testing strategies on the simulator to reduce total tests and identify outbreak clusters.
- Developed a web application to optimally allocate collected samples across testing centers based on payload and geographical constraints.

Selected Projects

🔗 Generating Wikipedia infoboxes using LLMs | LLMs, Vector Databases, Natural Language Processing Dec 2023

- Developed an LLM pipeline to generate tabular summaries of Wikipedia pages over existing manual process.
- Scraped 2k+ articles to build a vector DB and proposed a novel similarity-based template retrieval system.
- Reported an improvement of 97% in BLEU scores over zero-shot prompting and comparable accuracy to ground-truth on downstream QA tasks.

🔗 Analysis of Langevin Algorithms | Diffusion Models, Sampling, Optimization May 2023

- Studied and implemented various algorithms for reverse diffusion in Diffusion models.
- Implemented a dimensional-independent version of Langevin Algorithm(ULA-PD) enabling dimension-free convergence.
- Proposed and evaluated a metropolis version of ULA-PD for faster convergence of neural network optimization.

🔗 Excess Food Distribution Application | MySQL, Flask, React Dec 2022

- Developed a web application using Flask and React linking users & restaurants for food redistribution and analytics.
- Implemented triggers, stored procedures, & indexing for superior system performance.

Technical Skills

Languages: Python, Java, C++, SQL

Technologies/Frameworks: PyTorch, MongoDB, Neo4j, MySQL, L^AT_EX, Flask, React